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Annexe 2

Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du
mètre et aux représentations secondaires de la
seconde

**Recommended values of standard frequencies
for applications including the practical
realization of the metre and secondary
representations of the second**

Partie 2

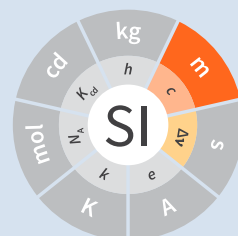
Part 2

Indium 115 Ion ()

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CCL-CCTF Frequency Standards Working Group

April 2018
avril 2018



**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 2: Indium 115 Ion ($^{115}\text{In}^+$)**

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1. Recommended value [1] of the frequency in the CIPM List of Frequencies

$$f(^{115}\text{In}^+) = 1\,267\,402\,452\,901\,050\text{ Hz}$$

equivalent to

$$\lambda(^{115}\text{In}^+) = 236\,540\,853.549\,544\text{ fm},$$

with a relative standard uncertainty of 1.6×10^{-14} .

2. Source data

Adopted value $f(^{115}\text{In}^+) = 1\,267\,402\,452\,901\,050\text{ Hz}$

with a relative uncertainty of 1.6×10^{-14} calculated from

$f(^{115}\text{In}^+)/\text{Hz}$	u/Hz	source data
1 267 402 452 899 920	690	[2] ^(a)
1 267 402 452 901 265	256	[3]
1 267 402 452 901 049.9	20.7	[4] ^(b)

- (a) From the least square procedure it turned out that the value and the uncertainty given in [2] is not compatible with the remaining data. Thus the uncertainty given in the original publication [2] was increased to 690 Hz to make it statistically more consistent.
- (b) The uncertainty of 6.9 Hz presented in ref. [4] was increased by a factor of three because it dominates the result with its uncertainty much lower than any of the others. Another fourth measurement [5] has not been used since the data were identical with the ones of ref. [3] but a much lower uncertainty of 63 Hz was reported without reasoning. The modification of the uncertainties followed the procedures outlined in ref. [6].

as a weighted mean of the above values.

$^{115}\text{In}^+$, $5s^2\ ^1\text{S}_0 - 5s5p\ ^3\text{P}_0$ unperturbed optical transition

References

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- [6] F. Riehle, P. Gill, F. Arias, L. Robertsson: The CIPM List of Recommended Frequency Standard Values: Guidelines and Procedures; *Metrologia* **55**, 188 – 200 (2018)